

## Semiconductors - Taiwan

## PO updates on semiconductor names after reflecting rate hike &amp; stronger cyclical

Price Objective Change

## PO cuts on eMemory and WT Micro

We update our POs for eMemory, WT Micro and GlobalWafers in this report, including trimming our POs for eMemory and WT Micro to NT\$4,500 (from NT\$5,500) and NT\$310 (from NT\$370) respectively after reflecting the impact from expected [US rate hikes \(see Morning Market Tidbits report\)](#), and raising our PO for GlobalWafers to NT\$685 (from NT\$575) along with an increase in our EPS estimates. For eMemory, our valuation is unchanged based on DCF, where we now assume a 7% WACC (vs 6% prior).

## Semiconductor IP: stay more constructive on eMemory

In our view, **eMemory** remains well-positioned to benefit from the IP (intellectual property) adoption by BMC (baseboard management controller), CPU (central processing unit), and controller/networking chips with Caliptra 1.0 standard from Vera Rubin platform. We also flag its business opportunity in the longer term via providing the OTP (one-time programmable) IP for SRAM (static random access memory) repair, given the potential adoption of SoIC (system on integrated chip) and CFET (complementary field effective transistor) structure, which counters the difficulty for SRAM shrinkage in more advanced nodes. On the other hand for **M31**, we see no meaningful progress of its 12nm project with key US IDM (integrated device manufacturer), which reinforces the uncertainty from higher revenue mix from licensing business. Further, its operating leverage should be limited in 2026 in our view given ~10% headcount increase year-to-date, while it seems to take a longer while for the benefits from ARM's exits in foundation IP market to materialize. Within the semiconductor IP sector, we reiterate Buy on eMemory and maintain Underperform on M31, in view of eMemory's steadier growth (vs greater uncertainty related to M31 given its 80+% revenue mix from licensing).

## Raise PO for GlobalWafers along with increase in estimates

We cut our 2026-28E EPS by 3% for **WT Micro**, as we reflect heavier interest expense due to higher expected interest rates, and our PO moves to NT\$310 (from NT\$370) now based on 10x (from 11x) 2H27-1H28E P/E given a softer earnings growth outlook. We reiterate our Buy rating on the stock as its datacenter business associated with TPU (tensor processing unit) remains on the uptrend into 2027-28E. For **GlobalWafers**, we raise our 2027/28E EPS by 13%/30%, after incorporating the stronger demand outlook in both memory and logic chips shared by [Micron \(see report\)](#) and the [increase in global semi forecasts \(see US Semiconductors report\)](#). Our PO for GlobalWafers goes to NT\$685 (from NT\$575), and is now based on 20x (was 19x) 2027E P/E, given the strengthening earnings growth outlook. However, we reiterate our Underperform rating on the stock given its stretched valuation at 39x/29x 2027/28E P/E.

**This research report provides general information only. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or other persons without the express written consent of BofA Securities.**

>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.

Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.

**BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.**

Refer to important disclosures on page 7 to 11. Analyst Certification on page 4. Price

07 July 2026

Equity  
Taiwan  
Semiconductors

**Mike Yang >>**  
Research Analyst  
Merrill Lynch (Taiwan)  
+886 2 2376 3729  
[mike.c.yang@bofa.com](mailto:mike.c.yang@bofa.com)

**Haas Liu >>**  
Research Analyst  
Merrill Lynch (Taiwan)  
+886 2 2376 3727  
[haas.liu@bofa.com](mailto:haas.liu@bofa.com)

**Cathy Hsu >>**  
Research Analyst  
Merrill Lynch (Taiwan)  
+886 2 2376 3726  
[cathy.hsu3@bofa.com](mailto:cathy.hsu3@bofa.com)

WACC – weighted average cost of capital

## Exhibit 1: Summary of PO changes

We adjust POs on three names in this report

| Company      | Ticker  | PO (local currency) |      |
|--------------|---------|---------------------|------|
|              |         | New                 | Old  |
| eMemory      | 3529 TT | 4500                | 5500 |
| WT Micro     | 3036 TT | 310                 | 370  |
| GlobalWafers | 6488 TT | 685                 | 575  |

Source: BofA Global Research

BofA GLOBAL RESEARCH

**Exhibit 2: Summary of changes in PO and valuation method**

We cut the target P/E multiple on WT Micro but raise the target P/E multiple on GlobalWafers in this report

| Company      | Ticker  | PO (local currency) |      | Valuation method   |                    |
|--------------|---------|---------------------|------|--------------------|--------------------|
|              |         | New                 | Old  | New                | Old                |
| eMemory      | 3529 TT | 4500                | 5500 | DCF (unchanged)    |                    |
| WT Micro     | 3036 TT | 310                 | 370  | 10x 2H27-1H28E P/E | 11x 2H27-1H28E P/E |
| GlobalWafers | 6488 TT | 685                 | 575  | 20x 2027E P/E      | 19x 2027E P/E      |

Source: BofA Global Research

BofA GLOBAL RESEARCH

**Exhibit 3: Table of recommendations**

Stocks mentioned with ratings in this report

| BofA ticker | Ticker  | Company name | Price (LC) | Rating |
|-------------|---------|--------------|------------|--------|
| XYLWF       | 3529 TT | eMemory      | NT\$ 2805  | C-1-7  |
| XMTZF       | 6643 TT | M31          | NT\$ 497   | C-3-7  |
| XZOPF       | 3036 TT | WT Micro     | NT\$ 219   | B-1-7  |
| XWLFF       | 6488 TT | GlobalWafers | NT\$ 1335  | C-3-7  |

Source: BofA Global Research, company data

BofA GLOBAL RESEARCH

**Exhibit 4: Earnings estimate changes – GlobalWafers**

We raise our 2027/28E EPS by 13%/30% after reflecting the assumption of stronger revenue growth and operating leverage

| NT\$b            | 2026E |       |         | 2027E |       |        | 2028E |       |        |
|------------------|-------|-------|---------|-------|-------|--------|-------|-------|--------|
|                  | New   | Old   | Diff    | New   | Old   | Diff   | New   | Old   | Diff   |
| Revenue          | 63.2  | 63.0  | 0%      | 82.2  | 74.4  | 11%    | 96.2  | 80.8  | 19%    |
| Gross profit     | 14.3  | 15.0  | -5%     | 25.5  | 20.6  | 24%    | 33.5  | 24.5  | 37%    |
| Gross margin     | 22.6% | 23.8% | -1.2ppt | 31.0% | 27.7% | 3.3ppt | 34.9% | 30.3% | 4.5ppt |
| Operating income | 8.0   | 8.7   | -8%     | 17.6  | 13.5  | 30%    | 24.5  | 16.9  | 45%    |
| Operating margin | 12.7% | 13.9% | -1.2ppt | 21.4% | 18.1% | 3.3ppt | 25.4% | 20.9% | 4.5ppt |
| Pretax income    | 10.8  | 11.6  | -6%     | 21.4  | 18.4  | 16%    | 29.0  | 21.8  | 33%    |
| Pretax margin    | 17.1% | 18.4% | -1.3ppt | 26.1% | 24.8% | 1.3ppt | 30.2% | 26.9% | 3.2ppt |
| Net income (A)   | 8.6   | 9.3   | -7%     | 16.5  | 14.5  | 13%    | 22.3  | 17.1  | 30%    |
| EPS (NT\$)       | 18.1  | 19.4  | -7%     | 34.4  | 30.4  | 13%    | 46.5  | 35.9  | 30%    |

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

**Exhibit 5: BofAe vs consensus – GlobalWafers**

We are 19%/12% ahead of consensus for 2027/28E EPS given the assumption of stronger gross margin and operating leverage

| NT\$b         | 2026E |           |         | 2027E |           |        | 2028E |           |        |
|---------------|-------|-----------|---------|-------|-----------|--------|-------|-----------|--------|
|               | BofA  | Consensus | Diff    | BofA  | Consensus | Diff   | BofA  | Consensus | Diff   |
| Revenue       | 63.2  | 65.3      | -3%     | 82.2  | 79.6      | 3%     | 96.2  | 92.4      | 4%     |
| Gross profit  | 14.3  | 15.6      | -9%     | 25.5  | 22.4      | 14%    | 33.5  | 31.5      | 6%     |
| Gross margin  | 22.6% | 23.9%     | -1.3ppt | 31.0% | 28.1%     | 2.9ppt | 34.9% | 34.1%     | 0.8ppt |
| Opt income    | 8.0   | 9.5       | -15%    | 17.6  | 16.0      | 10%    | 24.5  | 22.8      | 7%     |
| Opt margin    | 12.7% | 14.5%     | -1.8ppt | 21.4% | 20.0%     | 1.3ppt | 25.4% | 24.7%     | 0.8ppt |
| Pretax income | 10.8  | 12.0      | -10%    | 21.4  | 16.8      | 28%    | 29.0  | 24.0      | 21%    |
| Pretax margin | 17.1% | 18.5%     | -1.3ppt | 26.1% | 21.1%     | 5.0ppt | 30.2% | 26.0%     | 4.2ppt |
| Net income    | 8.6   | 9.7       | -11%    | 16.5  | 13.9      | 19%    | 22.3  | 19.8      | 12%    |
| EPS (NT\$)    | 18.1  | 20.3      | -11%    | 34.4  | 29.0      | 19%    | 46.5  | 41.5      | 12%    |

Source: BofA Global Research estimates, Bloomberg

BofA GLOBAL RESEARCH



**Exhibit 6: Income statement – GlobalWafers**

We expect the company's revenue to grow by 20+% CAGR during 2026-28E

| price                      | 1Q26     | 2Q26E    | 3Q26E    | 4Q26E    | 1Q27E    | 2Q27E    | 3Q27E    | 4Q27E    | 2025     | 2026E    | 2027E    | 2028E    |
|----------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Revenue                    | 13,985   | 15,382   | 16,424   | 17,454   | 18,521   | 19,794   | 21,169   | 22,698   | 60,598   | 63,245   | 82,183   | 96,176   |
| Cost of Revenue            | (11,062) | (11,856) | (12,942) | (13,111) | (13,460) | (13,913) | (14,484) | (14,844) | (45,974) | (48,972) | (56,700) | (62,638) |
| Gross profit               | 2,923    | 3,525    | 3,482    | 4,343    | 5,061    | 5,881    | 6,686    | 7,855    | 14,624   | 14,273   | 25,482   | 33,538   |
| Operating exp              | (1,439)  | (1,506)  | (1,605)  | (1,718)  | (1,798)  | (1,915)  | (2,030)  | (2,156)  | (5,987)  | (6,268)  | (7,899)  | (9,067)  |
| Operating income           | 1,484    | 2,020    | 1,877    | 2,624    | 3,263    | 3,966    | 4,655    | 5,698    | 8,636    | 8,005    | 17,583   | 24,471   |
| Non-opt net                | 872      | 552      | 555      | 859      | 947      | 988      | 932      | 988      | 880      | 2,838    | 3,855    | 4,542    |
| Income before tax          | 2,356    | 2,571    | 2,432    | 3,483    | 4,210    | 4,954    | 5,588    | 6,687    | 9,516    | 10,843   | 21,438   | 29,013   |
| Income Tax                 | (452)    | (643)    | (242)    | (871)    | (1,052)  | (1,238)  | (1,006)  | (1,672)  | (2,205)  | (2,208)  | (4,968)  | (6,758)  |
| Net income                 | 1,904    | 1,929    | 2,189    | 2,613    | 3,157    | 3,715    | 4,582    | 5,015    | 7,312    | 8,634    | 16,470   | 22,255   |
| Diluted EPS (NT\$)         | 3.98     | 4.03     | 4.58     | 5.46     | 6.60     | 7.77     | 9.58     | 10.49    | 15.3     | 18.1     | 34.4     | 46.5     |
| <b>Margin; tax rate</b>    |          |          |          |          |          |          |          |          |          |          |          |          |
| Gross margin               | 20.9%    | 22.9%    | 21.2%    | 24.9%    | 27.3%    | 29.7%    | 31.6%    | 34.6%    | 24.1%    | 22.6%    | 31.0%    | 34.9%    |
| Operating expenses         | 10.3%    | 9.8%     | 9.8%     | 9.8%     | 9.7%     | 9.7%     | 9.6%     | 9.5%     | 9.9%     | 9.9%     | 9.6%     | 9.4%     |
| Operating margin           | 10.6%    | 13.1%    | 11.4%    | 15.0%    | 17.6%    | 20.0%    | 22.0%    | 25.1%    | 14.3%    | 12.7%    | 21.4%    | 25.4%    |
| Pretax margin              | 16.8%    | 16.7%    | 14.8%    | 20.0%    | 22.7%    | 25.0%    | 26.4%    | 29.5%    | 15.7%    | 17.1%    | 26.1%    | 30.2%    |
| Net margin                 | 13.6%    | 12.5%    | 13.3%    | 15.0%    | 17.0%    | 18.8%    | 21.6%    | 22.1%    | 12.1%    | 13.7%    | 20.0%    | 23.1%    |
| Tax rate                   | 23.6%    | 25.4%    | 20.8%    | 23.2%    | 20.4%    | 23.2%    | 0.0%     | 0.0%     | 23.2%    | 20.4%    | 23.2%    | 23.3%    |
| <b>Sequential growth %</b> |          |          |          |          |          |          |          |          |          |          |          |          |
| Revenue                    | -3.6%    | 10.0%    | 6.8%     | 6.3%     | 6.1%     | 6.9%     | 6.9%     | 7.2%     | -3.2%    | 4.4%     | 29.9%    | 17.0%    |
| Gross profit               | -21.5%   | 20.6%    | -1.2%    | 24.7%    | 16.5%    | 16.2%    | 13.7%    | 17.5%    | -26.2%   | -2.4%    | 78.5%    | 31.6%    |
| Operating exp              | 6.8%     | 4.6%     | 6.6%     | 7.1%     | 4.6%     | 6.5%     | 6.0%     | 6.2%     | 5.3%     | 4.7%     | 26.0%    | 14.8%    |
| Operating income           | -37.6%   | 36.1%    | -7.1%    | 39.9%    | 24.3%    | 21.5%    | 17.4%    | 22.4%    | -38.8%   | -7.3%    | 119.7%   | 39.2%    |
| Net income                 | -13.6%   | 1.3%     | 13.5%    | 19.3%    | 20.9%    | 17.7%    | 23.3%    | 9.5%     | -25.7%   | 18.1%    | 90.7%    | 35.1%    |

Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

**Price objective basis & risk****eMemory (XYLWF)**

Our PO of NT\$4,500 is based on a discounted cash flow (DCF) approach, with weighted average cost of capital (WACC) 7% and terminal growth rate of 3%.

Downside risks are:

- (1) wafer price drop due foundry competition/over-capacity, especially in 28nm and more advanced process, which could curtail the revenue incurred from royalty business for eMemory
- (2) weaker-than-expected node migration, which may slow down the adoption increase of NeoFuse technology offered by eMemory
- (3) slower-than-expected IP development due to bottleneck or challenge in talent acquisition, which may cause negative impact to the company's long-term growth potential
- (4) other types of memory emerge to replace key solutions/IPs offered by eMemory.

**GlobalWafers (XWLFF)**

Our PO of NT\$685 is based on 20x 2027E average P/E, at the mid-end of the company's trading history (2016-25), which we consider fair given its No.3 position in silicon wafer market globally (with c.17% market share) and macro uncertainties.

Downside risks to our PO are:

- 1) Unexpected capacity expansion by top-5 players
- 2) faster-than-expected breakthrough by emerging Chinese suppliers
- 3) significant shortfall in global semi demand.

Upside risks to our PO are:

- 1) further strengthening supply and demand situation to benefit wafer pricing and gross



margin uptick

2) sustained recovery in global semi demand

### **M31 Technology (XMTZF)**

We base our PO NT\$435 on 50x 2H26-1H27E P/E, which is at around low-end to mid of the company's historical trading range. In our view, the valuation is backed by 1) 76% EPS CAGR during 2026-28E (implying 0.7x P/E to growth ratio), 2) improving ROE to 18% in 2027-28E, and 3) recovering operating margin to 18% in 2027-28E.

Upside risks are 1) a smoother-than-expected progress in recognizing the licensing revenue from different foundries' platforms, 2) a recovering momentum in the design activity by chip-design customers.

Downside risks are 1) the downside surprise from semiconductor market demand and foundries' utilization rates, 2) keener competition and difficulties in attracting and/or retaining the talents.

### **WT Microelectronics (XZOPF)**

We base our PO NT\$310 on 10x 2H27-1H28E P/E, which is at the mid-end of its historical trading range of 4-15x. This is justified in our view, given the support from: 1) an operating margin of 2.0-2.1% in 2027-28E, 2) improving ROE toward 26% in 2027-28E (vs 12% in 2024-25), and 3) 22% EPS CAGR during 2026-28E.

Upside risks to our PO are:

- 1) faster-than-expected demand recovery in auto/industrial verticals, especially from US/Europe market
- 2) a lower interest rate that could alleviate the company's financial burden at non-operating level.

Downside risks to our PO are:

- 1) a revenue decline in datacenter/server related business, which has been the key growth driver during 2021-24
- 2) market share loss as suppliers or customers seek other distributors' help after WT Micro consolidates Future Electronics.
- 3) Potential overhang from share dilution.

## **Analyst Certification**

I, Mike Yang, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

## **Special Disclosures**

Information on securities which are listed on the exchanges where ML Securities (Taiwan) Limited is not permitted to trade or solicit trades for clients is for informational purposes only and is not a recommendation or a solicitation to trade such securities. ML Securities (Taiwan) Limited will not execute transactions for nor accept orders from clients to trade such securities. Foreign investment in Taiwan securities is regulated and restricted. Currently, foreign investment in Taiwan securities is permitted by investment through: (1) global depository receipts, (2) convertible bonds, (3) mutual funds issued offshore of Taiwan, and (4) a special foreign institutional investors (FINIs) and foreign individual investors (FIDIs) program supervised by the Taiwan SFB whereunder FINIs/FIDIs may apply for investment ID to invest in Taiwan securities by registration with Taiwan Stock Exchange. FINIs will additionally need consent from the foreign exchange authority,

ie, the Central Bank of China. In addition to the limitations above, various industry-specific percentage-based limitations on foreign ownership of Taiwan companies (and in some cases prohibitions) may apply. Investments are subject to exchange rate and currency conversion restrictions and risks. Dividends and interest earned by foreign investors' Taiwan securities/instruments are generally subject to a 20% withholding tax. Ordinary shares are not available to ML private client accounts in the U.S.

This report is distributed in Taiwan by Merrill Lynch Securities (Taiwan) Limited, which is regulated by the Taiwan SFB.

BofA Securities is currently acting as financial advisor to One Investment Management Ltd in connection with the proposed acquisition with PayPay Corporation, a subsidiary of SoftBank Group Corp., of shares of T&D Financial Life Insurance Company, a consolidated subsidiary of T&D Holdings, Inc., which was announced on June 4, 2026.

BofA Securities is currently acting as joint financial advisor to ABB Ltd in connection with the sale of its Robotics Division to SoftBank Group Corp, which was announced on October 8, 2025.

**APR - Semiconductor Coverage Cluster**

| Investment rating   | Company                                  | BofA Ticker | Bloomberg symbol | Analyst        |
|---------------------|------------------------------------------|-------------|------------------|----------------|
| <b>BUY</b>          |                                          |             |                  |                |
|                     | Alchip                                   | ALCPF       | 3661 TT          | Haas Liu       |
|                     | AMEC                                     | XRTOF       | 688012 CH        | Dai Shen       |
|                     | ASE Technology Holding                   | XSRIF       | 3711 TT          | Haas Liu       |
|                     | ASE Technology Holding -ADR              | ASX         | ASX US           | Haas Liu       |
|                     | ASMPT                                    | ASMVF       | 522 HK           | Simon Woo, CFA |
|                     | Aspeed                                   | XLKMF       | 5274 TT          | Mike Yang      |
|                     | Chroma ATE                               | CRMJF       | 2360 TT          | Haas Liu       |
|                     | eMemory                                  | XYLWF       | 3529 TT          | Mike Yang      |
|                     | GigaDevice                               | XGXIF       | 603986 CH        | Daley Li, CFA  |
|                     | Global Unichip Corp.                     | GBUHF       | 3443 TT          | Haas Liu       |
|                     | Grand Process Technology Corp            | XZOWF       | 3131 TT          | Haas Liu       |
|                     | Hanmi Semiconductor                      | HNSIF       | 042700 KS        | Simon Woo, CFA |
|                     | Hon Precision                            | XXPIF       | 7769 TT          | Haas Liu       |
|                     | Horizon Robotics                         | HRZRF       | 9660 HK          | Daley Li, CFA  |
|                     | Ingenic                                  | XISCF       | 300223 CH        | Dai Shen       |
|                     | InnoScience Technology                   | XSCHF       | 2577 HK          | Daley Li, CFA  |
|                     | JCET Group Co Ltd                        | XJIEF       | 600584 CH        | Dai Shen       |
|                     | King Yuan Electronics Corp.              | KYUFF       | 2449 TT          | Haas Liu       |
|                     | Kinik                                    | KIKCF       | 1560 TT          | Haas Liu       |
|                     | MediaTek                                 | MDTKF       | 2454 TT          | Haas Liu       |
|                     | Montage Technology                       | XRDF        | 688008 CH        | Daley Li, CFA  |
|                     | MPI Corporation                          | XMJCF       | 6223 TT          | Mike Yang      |
|                     | Nanya Technology                         | NNYAF       | 2408 TT          | Simon Woo, CFA |
|                     | NCE Power                                | XVFFF       | 605111 CH        | Daley Li, CFA  |
|                     | OmniVision                               | XXHQF       | 603501 CH        | Dai Shen       |
|                     | Phison Electronics                       | PISNF       | 8299 TT          | Simon Woo, CFA |
|                     | Powerchip Semiconductor Manufacturing Co | XCHPF       | 6770 TT          | Haas Liu       |
|                     | Powertech Technology                     | XPPZF       | 6239 TT          | Simon Woo, CFA |
|                     | Rockchip                                 | XRPXF       | 603893 CH        | Daley Li, CFA  |
|                     | Samsung C&T                              | XCDTF       | 028260 KS        | Simon Woo, CFA |
|                     | Samsung Elec -G                          | SSNHZ       | SMSN LI          | Simon Woo, CFA |
|                     | Samsung Electronics                      | SSNLF       | 005930 KS        | Simon Woo, CFA |
|                     | Samsung Electronics Preferred            | SSNPF       | 005935 KS        | Simon Woo, CFA |
|                     | Silergy Corp.                            | SLEGF       | 6415 TT          | Mike Yang      |
|                     | Silicon Motion                           | SIMO        | SIMO US          | Simon Woo, CFA |
|                     | SK Square                                | SKSQF       | 402340 KS        | Simon Woo, CFA |
|                     | Taiwan Semiconductor Manufacturing Co.   | TSM         | TSM US           | Haas Liu       |
|                     | Taiwan Semiconductor Manufacturing Co.   | TSMWF       | 2330 TT          | Haas Liu       |
|                     | Vanguard International Semiconductor Co  | VGILF       | 5347 TT          | Haas Liu       |
|                     | Winbond Electronics                      | WBEKF       | 2344 TT          | Dai Shen       |
|                     | WinWay Technology                        | XWCLF       | 6515 TT          | Mike Yang      |
|                     | WT Microelectronics                      | XZOPF       | 3036 TT          | Mike Yang      |
| <b>NEUTRAL</b>      |                                          |             |                  |                |
|                     | ASMedia Technology Inc.                  | XZSFF       | 5269 TT          | Mike Yang      |
|                     | Black Sesame Intl Holding                | BSIHF       | 2533 HK          | Daley Li, CFA  |
|                     | Crystal Clear                            | XPPTF       | 300655 CH        | Dai Shen       |
|                     | Hangzhou Silan Microelectronics          | XDFRF       | 600460 CH        | Daley Li, CFA  |
|                     | LX Semicon                               | XLXSF       | 108320 KS        | Simon Woo, CFA |
|                     | Novatek                                  | NVKMF       | 3034 TT          | Haas Liu       |
|                     | Realtek                                  | RLTKF       | 2379 TT          | Mike Yang      |
|                     | Shenzhen Goodix                          | XQPLF       | 603160 CH        | Daley Li, CFA  |
| <b>UNDERPERFORM</b> |                                          |             |                  |                |
|                     | Faraday                                  | FDYTF       | 3035 TT          | Haas Liu       |
|                     | GlobalWafers                             | XWLFF       | 6488 TT          | Mike Yang      |
|                     | Hua Hong Semi                            | HHUSF       | 1347 HK          | Dai Shen       |
|                     | Lion Electronics                         | XDHFF       | 605358 CH        | Dai Shen       |
|                     | M31 Technology                           | XMTZF       | 6643 TT          | Mike Yang      |
|                     | Macronix International                   | MXICF       | 2337 TT          | Dai Shen       |
|                     | Maxscend                                 | XMVSF       | 300782 CH        | Daley Li, CFA  |
|                     | Parade                                   | PRDWF       | 4966 TT          | Mike Yang      |
|                     | Soulbrain                                | XSBOF       | 357780 KS        | Simon Woo, CFA |
|                     | United Microelectronics Corp.            | XUMIF       | 2303 TT          | Haas Liu       |
|                     | United Microelectronics Corp.            | UMC         | UMC US           | Haas Liu       |

APR - Semiconductor Coverage Cluster

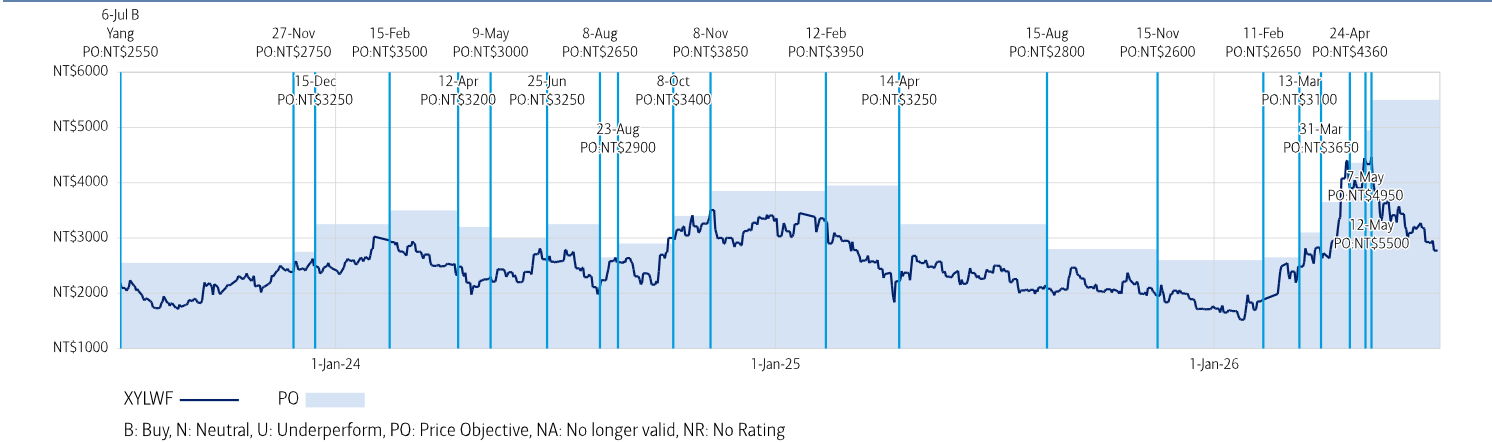
| Investment rating | Company            | BofA Ticker | Bloomberg symbol | Analyst        |
|-------------------|--------------------|-------------|------------------|----------------|
|                   | VeriSilicon        | XMLZF       | 688521 CH        | Daley Li, CFA  |
|                   | Win Semiconductors | WSMIF       | 3105 TT          | Haas Liu       |
|                   | Wonik IPS          | XRHQF       | 240810 KS        | Simon Woo, CFA |

| RSTR |          |       |           |                |
|------|----------|-------|-----------|----------------|
|      | SK Hynix | HXSCF | 000660 KS | Simon Woo, CFA |

Disclosures

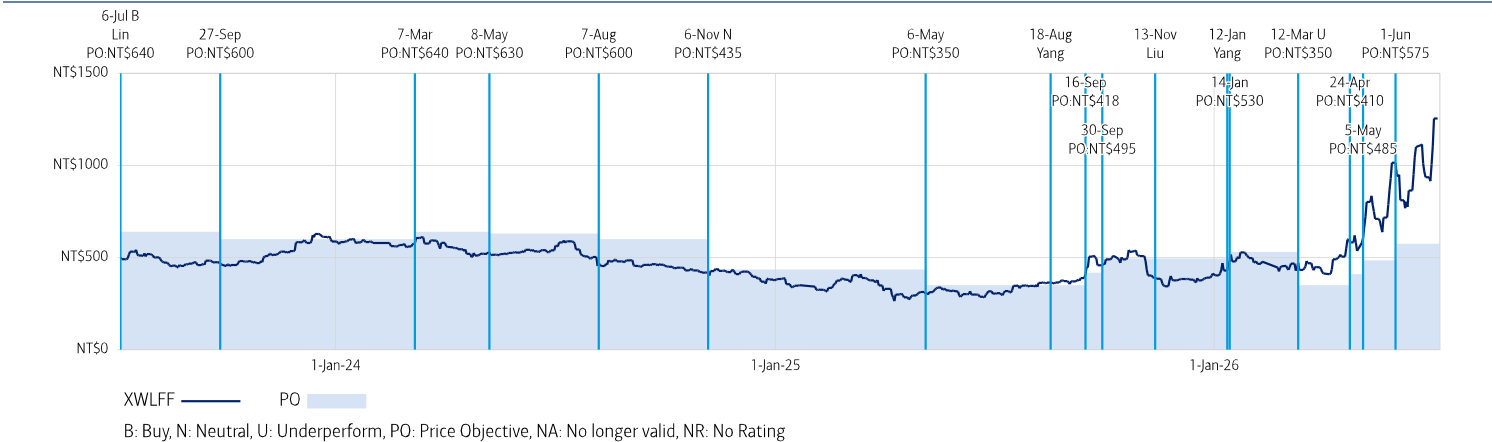
Important Disclosures

eMemory (XYLWF) Price Chart

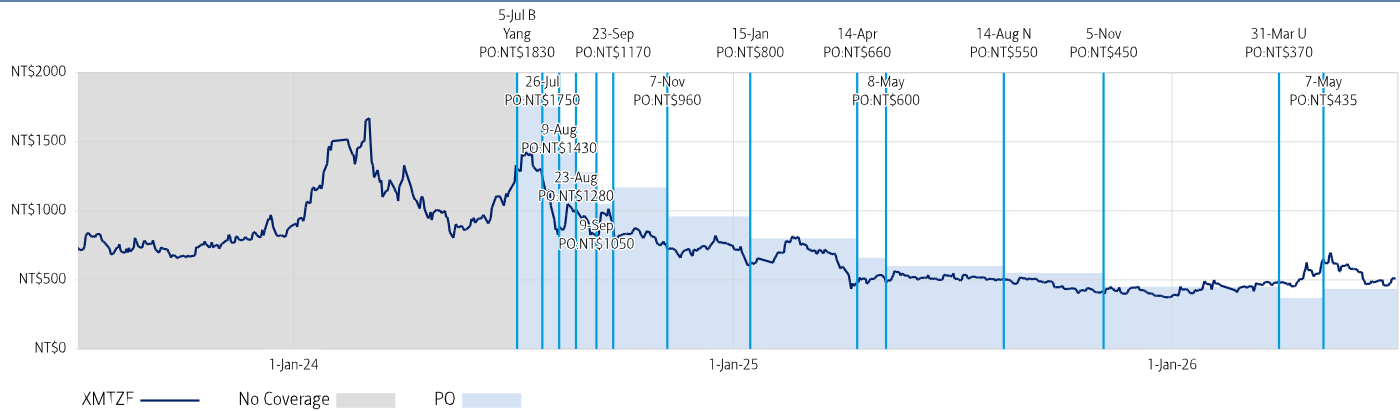


The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

GlobalWafers (XWLFF) Price Chart

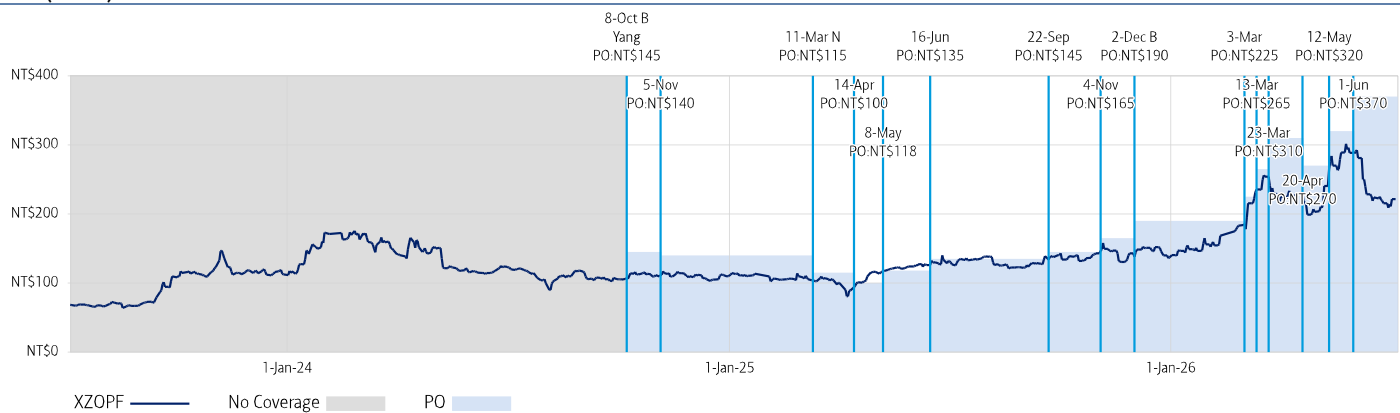


The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

**M31 Technology (XMTZF) Price Chart**

B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

**WT Micro (XZOPF) Price Chart**

B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

**Equity Investment Rating Distribution: Technology Group (as of 30 Jun 2026)**

| Coverage Universe | Count | Percent | Inv. Banking Relationships <sup>R1</sup> | Count | Percent |
|-------------------|-------|---------|------------------------------------------|-------|---------|
| Buy               | 243   | 60.60%  | Buy                                      | 123   | 50.62%  |
| Hold              | 87    | 21.70%  | Hold                                     | 45    | 51.72%  |
| Sell              | 71    | 17.71%  | Sell                                     | 21    | 29.58%  |

**Equity Investment Rating Distribution: Global Group (as of 30 Jun 2026)**

| Coverage Universe | Count | Percent | Inv. Banking Relationships <sup>R1</sup> | Count | Percent |
|-------------------|-------|---------|------------------------------------------|-------|---------|
| Buy               | 1987  | 56.11%  | Buy                                      | 1190  | 59.89%  |
| Hold              | 797   | 22.51%  | Hold                                     | 496   | 62.23%  |
| Sell              | 757   | 21.38%  | Sell                                     | 391   | 51.65%  |

<sup>R1</sup> Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.



**FUNDAMENTAL EQUITY OPINION KEY:** Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. **VOLATILITY RISK RATINGS**, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. **INVESTMENT RATINGS** reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

| Investment rating | Total return expectation (within 12-month period of date of initial rating) | Ratings dispersion guidelines for coverage cluster <sup>R2</sup> |
|-------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------|
| Buy               | ≥ 10%                                                                       | ≤ 70%                                                            |
| Neutral           | ≥ 0%                                                                        | ≤ 30%                                                            |
| Underperform      | N/A                                                                         | ≥ 20%                                                            |

<sup>R2</sup>Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

**INCOME RATINGS**, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. *Coverage Cluster* is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

Price Charts for the securities referenced in this research report are available on the [Price Charts website](#), or call 1-800-MERRILL to have them mailed.

BofA or an affiliate has received compensation from the issuer for non-investment banking services or products within the past 12 months: GlobalWafers, WT Micro.

The issuer is or was, within the last 12 months, a non-securities business client of BofA and/or one or more of its affiliates: GlobalWafers, WT Micro.

In the US, retail sales and/or distribution of this report may be made only in states where these securities are exempt from registration or have been qualified for sale: eMemory, GlobalWafers, M31 Technology, WT Micro.

The country in which this issuer is organized has certain laws or regulations that limit or restrict ownership of the issuer's shares by nationals of other countries: eMemory, GlobalWafers, M31 Technology, WT Micro.

The issuer is or was, within the last 12 months, a securities business client (non-investment banking) of BofA and/or one or more of its affiliates: GlobalWafers, WT Micro.

BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

## Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.

Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.

The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.

Officers of BofA or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.

Refer to [BofA Global Research policies relating to conflicts of interest](#).

**"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.**

### Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:

BofA and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at [www.bofam.com/BofASEdisclaimer](http://www.bofam.com/BofASEdisclaimer); BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSF); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority. This information: has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSF; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore)

(Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBL. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the [Electronic Communications Disclaimers](#) for further information.

This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

#### General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.

This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are largely unregulated. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities that may have been recommended in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments discussed in this report.

BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.

In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

#### Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQDatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.

Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.

Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.

The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.

All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.

Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BofA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.

Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.

In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.

Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.